Epidemiology of Airborne and DROPLET INFECTIONS

Dr Tasneem Mohammed Ali

AIR BORN & DROPLET INFECTION

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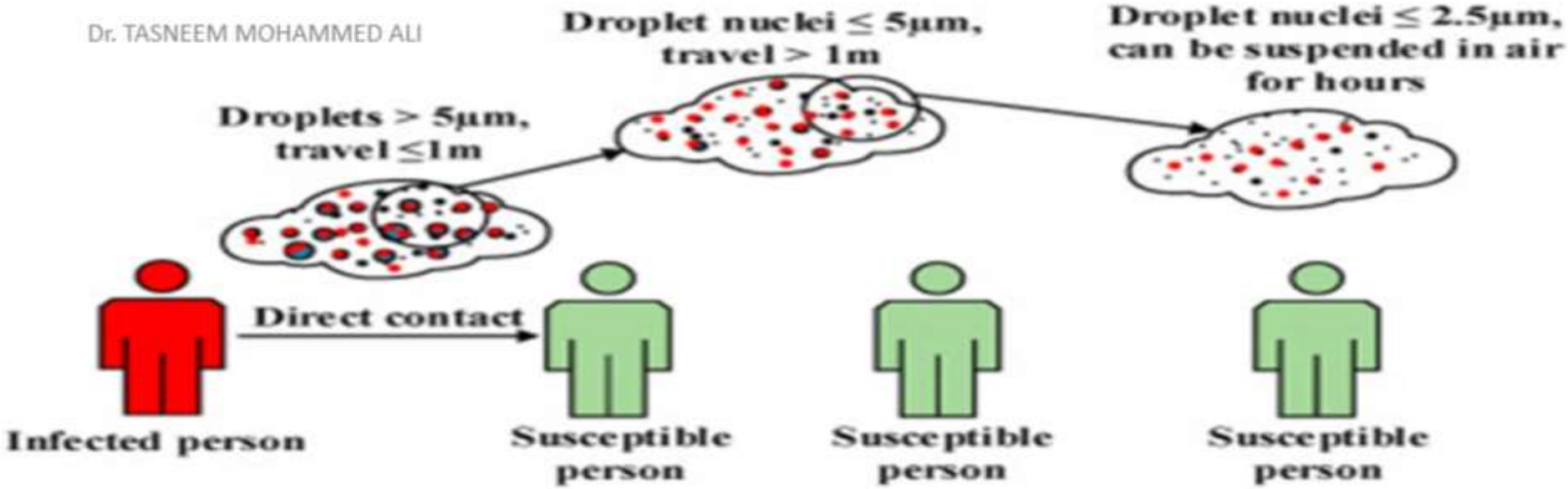

- It is a group of diseases in which a causative organism finds **entry** to the susceptible individuals through **upper respiratory tract.**

➢ Droplet refers to the very fine particles composed mainly of water and saliva that come out from mouth and nose during coughing, sneezing or talking

TRANSMISSION

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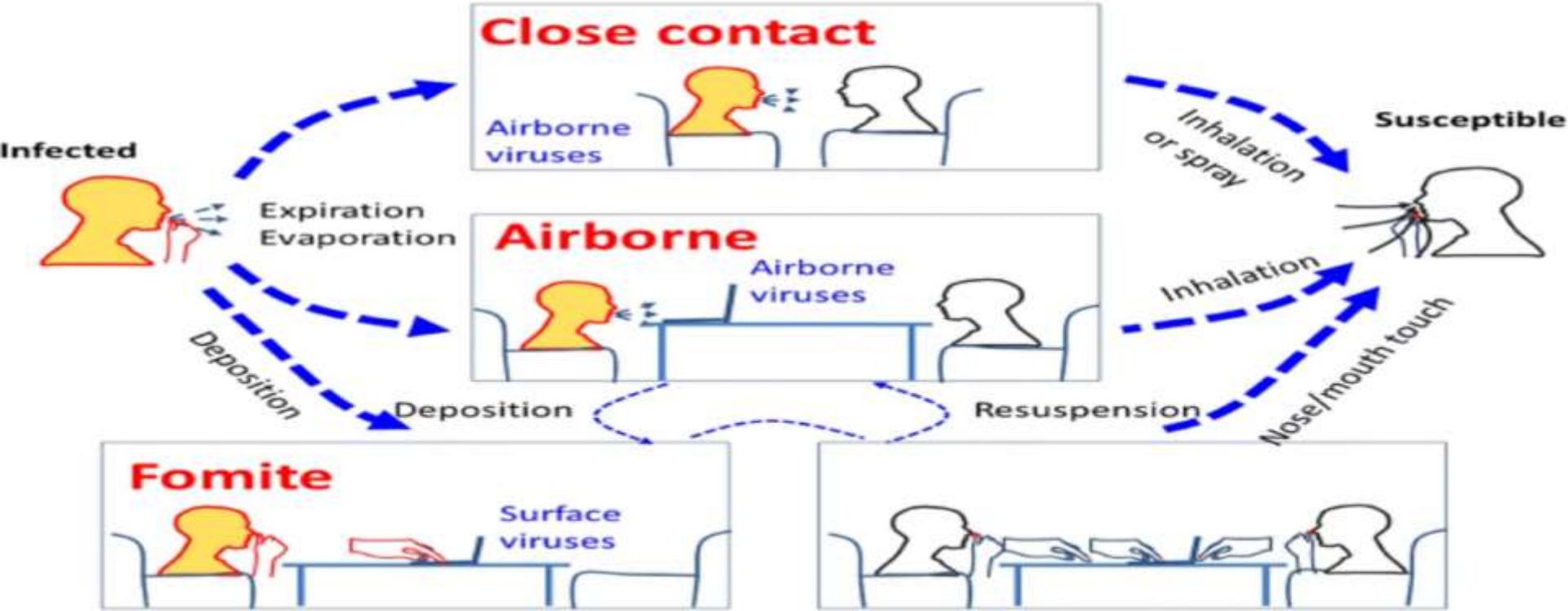
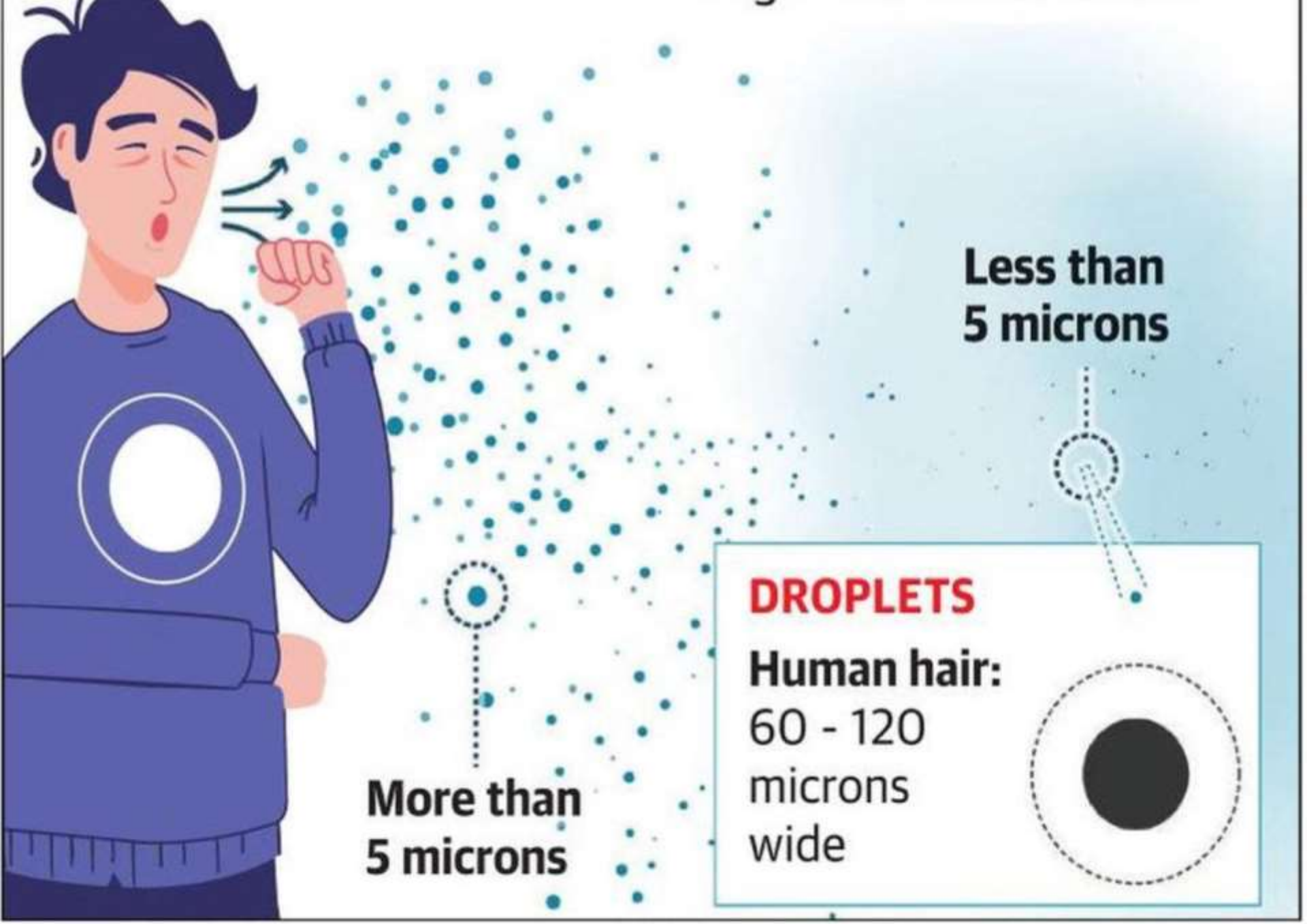
Direct droplet; the reservoir and susceptible are ***close*** to each other.

Indirect droplet; through **contaminated articles and foments**

Air borne infection; in case of fine droplet, by inhalation of suspended Particles in air, while in case of large droplet, they settle down and contaminate dust





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SOURCE: WORLD HEALTH ORGANIZATION

Classification of Droplet Infections:

A. Viral droplet infections:

1 Measles
2 German measles
3 Mumps
4 Chickenpox and Herpes Zoster
5 Smallpox
6. Influenza

B. **bacterial droplet infections:**

1. Whooping cough.
2. Meningococcal Meningitis.
3- Diphtheria.
4- Streptococcal Diseases.
5- Tuberculosis.

General Prevention and Control Measures of Droplet Infections

Ventilation :

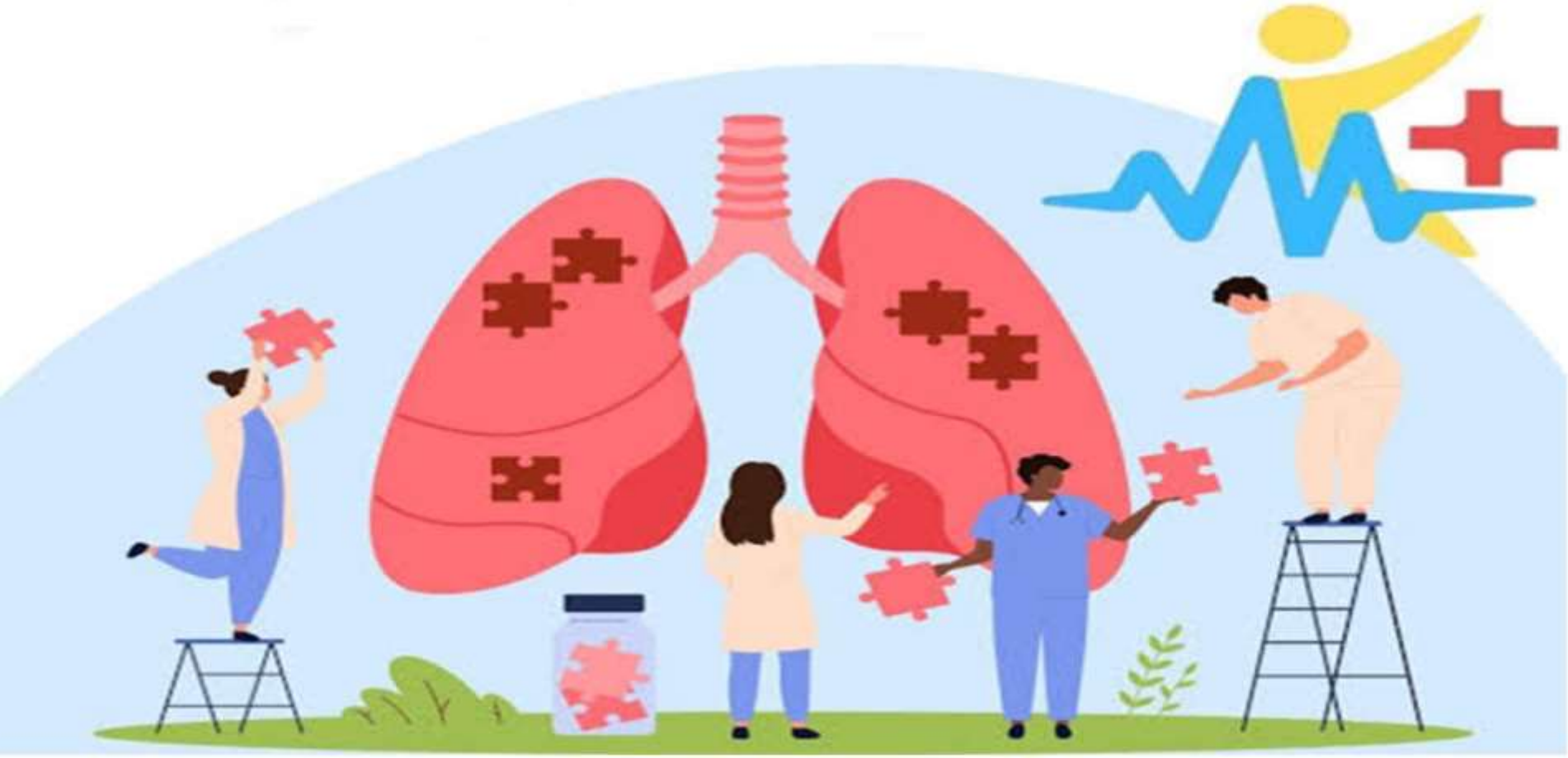
Since organisms transmissible through the air can be widely dispersed, **specific air ventilation** is required to manage their dispersion thus control outbreaks of airborne infections.

In routine areas of work and stay **adequacy of ventilation** at all times is ensured by provision of at least two windows per room with an area of about 10 percent of the floor space and arranged so as to provide cross ventilation.

Avoid Overcrowding : Other important measure, for preventing airborne infections is prevention of overcrowding. The minimum floor area should be 6 m2 per person. Beds must be spaced with an interval of at least two meters between the centres of the two adjacent beds.

Respiratory Protection Devices : Surgical masks that cover the mouth and nose should be worn by healthcare staff in hospitals and by the patient themselves also. N95 respirator /surgical mask are currently recommended.

TUBERCULOSIS



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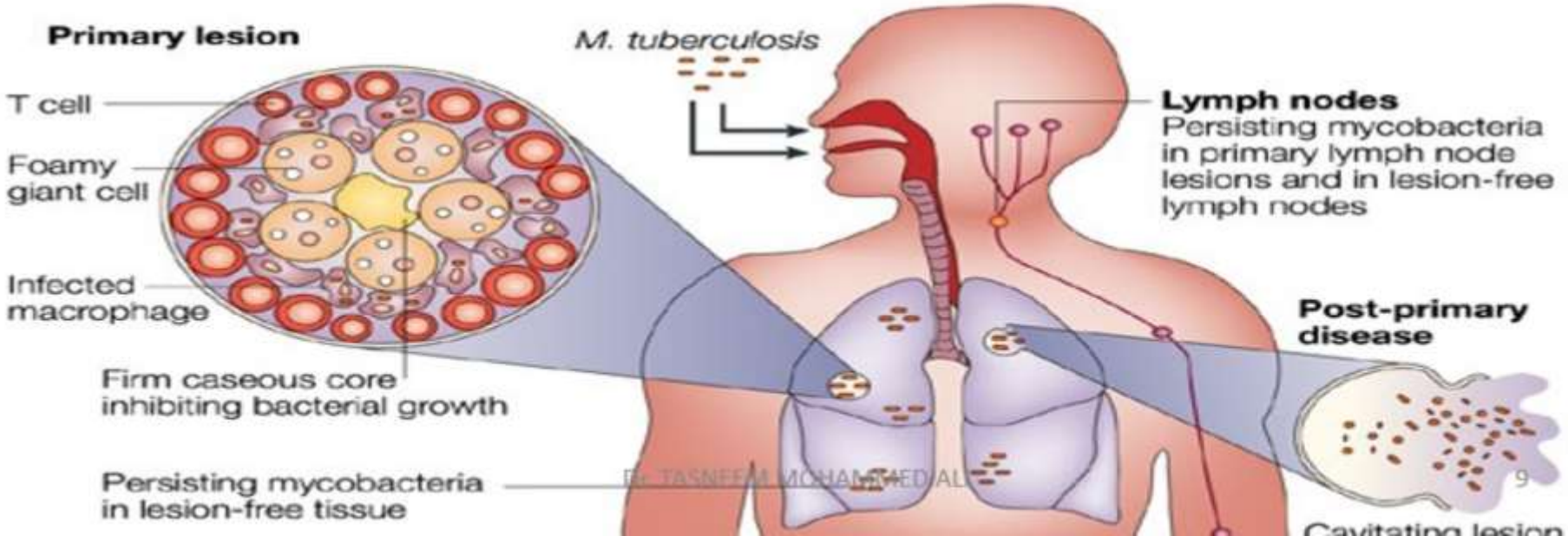
Tuberculosis

Tuberculosis is a worldwide, chronic communicable bacterial disease.

Tuberculosis disease is the outcome of the fight between virulence of the organism and resistance of the body.



Historical Perspective of Tuberculosis

9,000 Years Ago
Tuberculosis discovered in mummies in Egypt

1882
Robert Koch discovers tuberculosis

1921
Calmette & Guerin develop first TB vaccine

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Introduction & Problem Statement

TB is a **mycobacterium** disease important cause of disability & death worldwide, but is **curable** & preventable

Kills 5,000 people a day – **2-3 million each year**

1/3 of world's population is **infected** with TB

Kills **more young women** than any other disease

Multidrug-resistant TB (MDR-TB) public health crisis & security threat

Infectious TB can spread to **10-15 persons** per year

Ending the TB epidemic by **2030** is among health targets of UN SDGs

Magnitude of Tuberculosis in Egypt

- **Egypt** is a **low** TB-prevalence country
- **TB is consi**dered an important public health problem.
- The **Ministry** of health & population (MOHP) has established the National tuberculosis control program in 1979.

WHO TB Report, 2022 for Egypt

Prevalence	28 per 100,000
Incidence	17 per 100,000
Mortality	0.59 per 100,000

According to **WHO Global TB Report, 2022**

For Egypt Prevalence	28 per 100,000
Incidence	17 per 100,000
Mortality	0.59 per 100,000

- **Less than 1%** of adult TB patients are HIV-positive

TUBERCULOSIS CASE DISTRIBUTION IN EGYPT

Pulmonary TB (smear-positive & smear-negative) accounts for **80–85%** of all TB cases.

Extra-pulmonary TB represents **15–20%** of total cases.

For every **new smear-positive pulmonary TB case**, there is usually **one additional smear-negative pulmonary or extra-pulmonary case** present.

Egypt is a **low HIV-prevalence country**.

Currently, **less than 1% of adult TB patients are HIV-positive**.

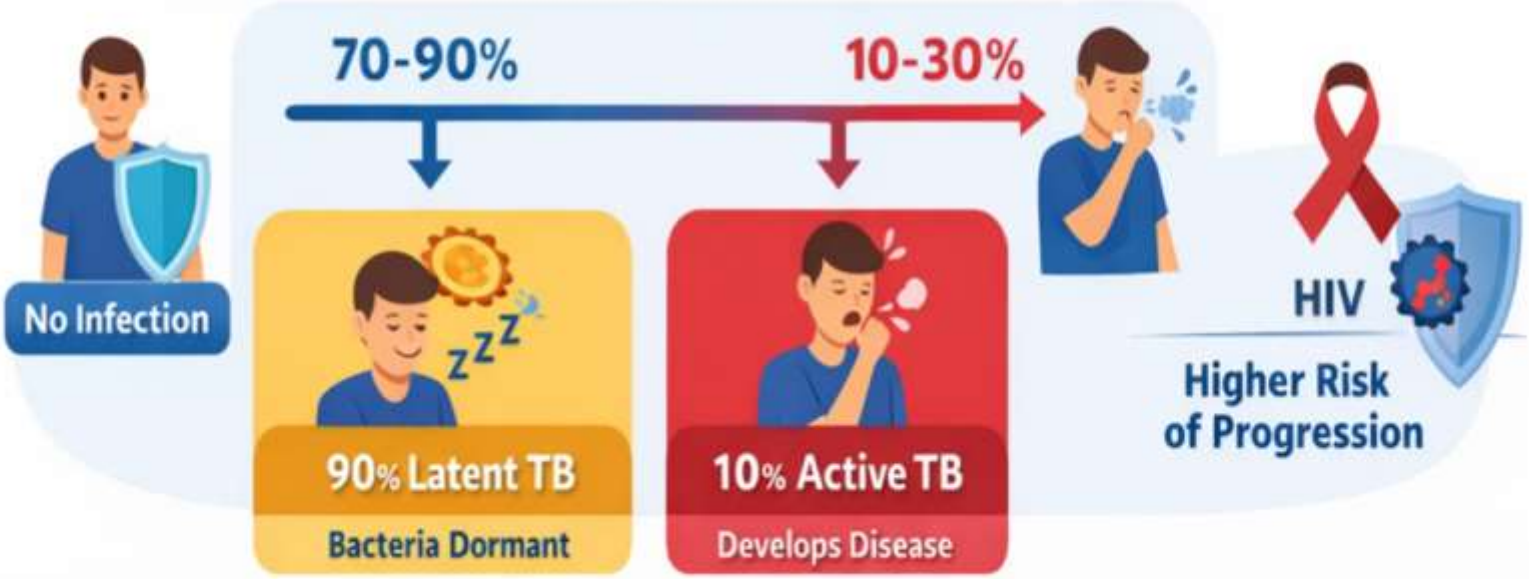
Continuous surveillance is essential because an increase in **HIV–TB co-infection** would complicate TB control efforts.

NATURAL HISTORY OF TUBERCULOSIS

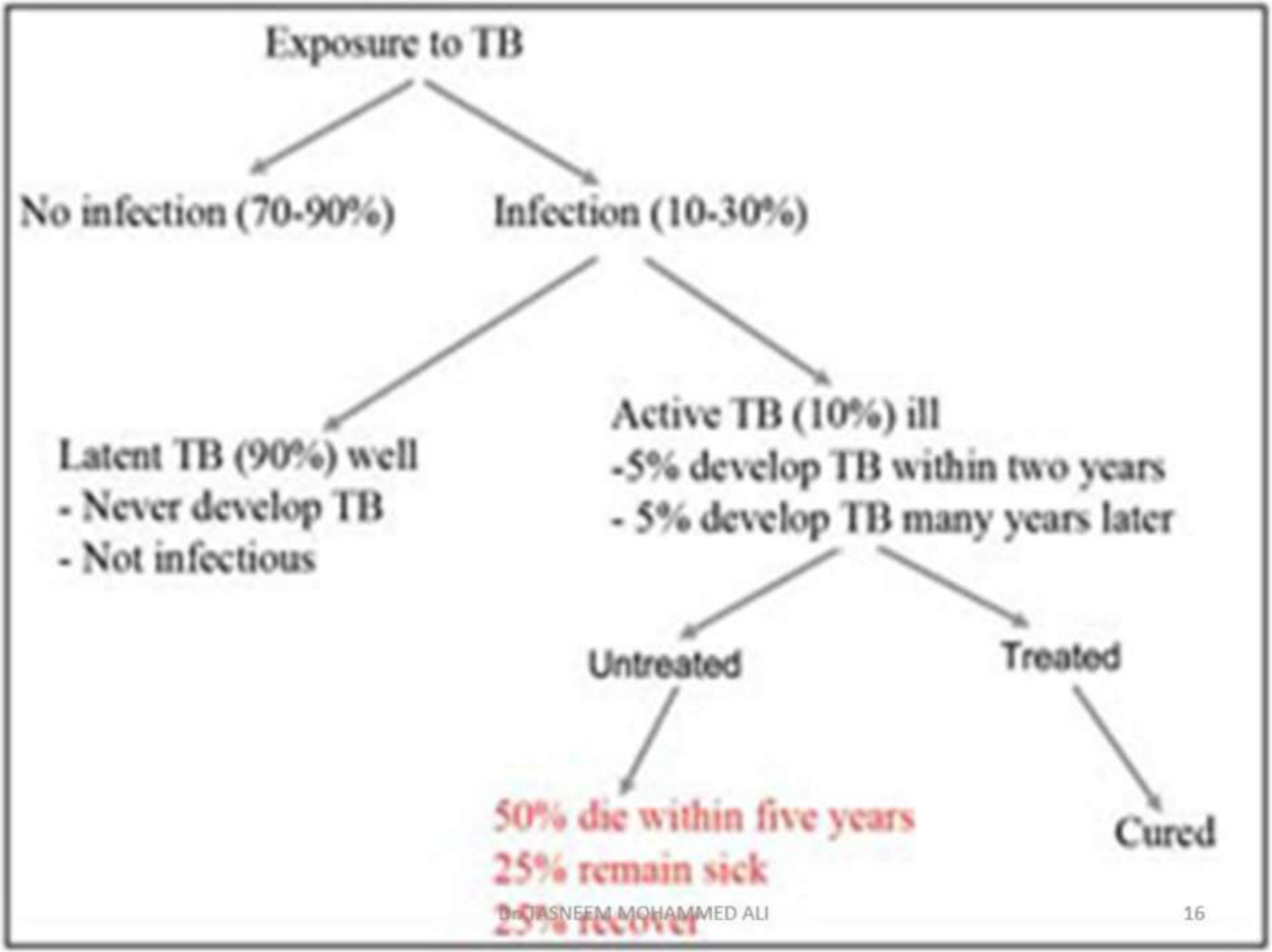
- **70–90%** of exposed individuals **do not become infected**.
- **10–30%** of exposed persons become infected with *Mycobacterium tuberculosis*.
- Of those infected:
  - **~90%** develop **Latent TB Infection (LTBI)** (immune system suppresses bacteria).
  - **~10%** develop **active TB disease** during their lifetime.
- In **HIV-infected persons**, progression from infection to active disease is **more rapid** due to weakened immunity.
- **Outcome of Untreated Pulmonary TB**
- **50%** die within 5 years (most deaths occur within 2 years).
- **25%** remain chronically ill and infectious (continue spreading TB).
- **25%** recover spontaneously due to strong immunity, but may relapse if bacteria remain latent.

Natural History of Tuberculosis

Exposure to TB



Progression Without Treatment


Exposure to TB
No infection (70-90%)
Infection (10-30%)
Latent TB (90%) well
- Never develop TB
- Not infectious
Active TB (10%) ill
-5% develop TB within two years
- 5% develop TB many years later
Untreated
Treated
50% die within five years
25% remain sick
25% recover
Cured

BASIC DEFINITION OF TUBERCULOSIS CASES (WHO 2013)

Presumptive TB Case

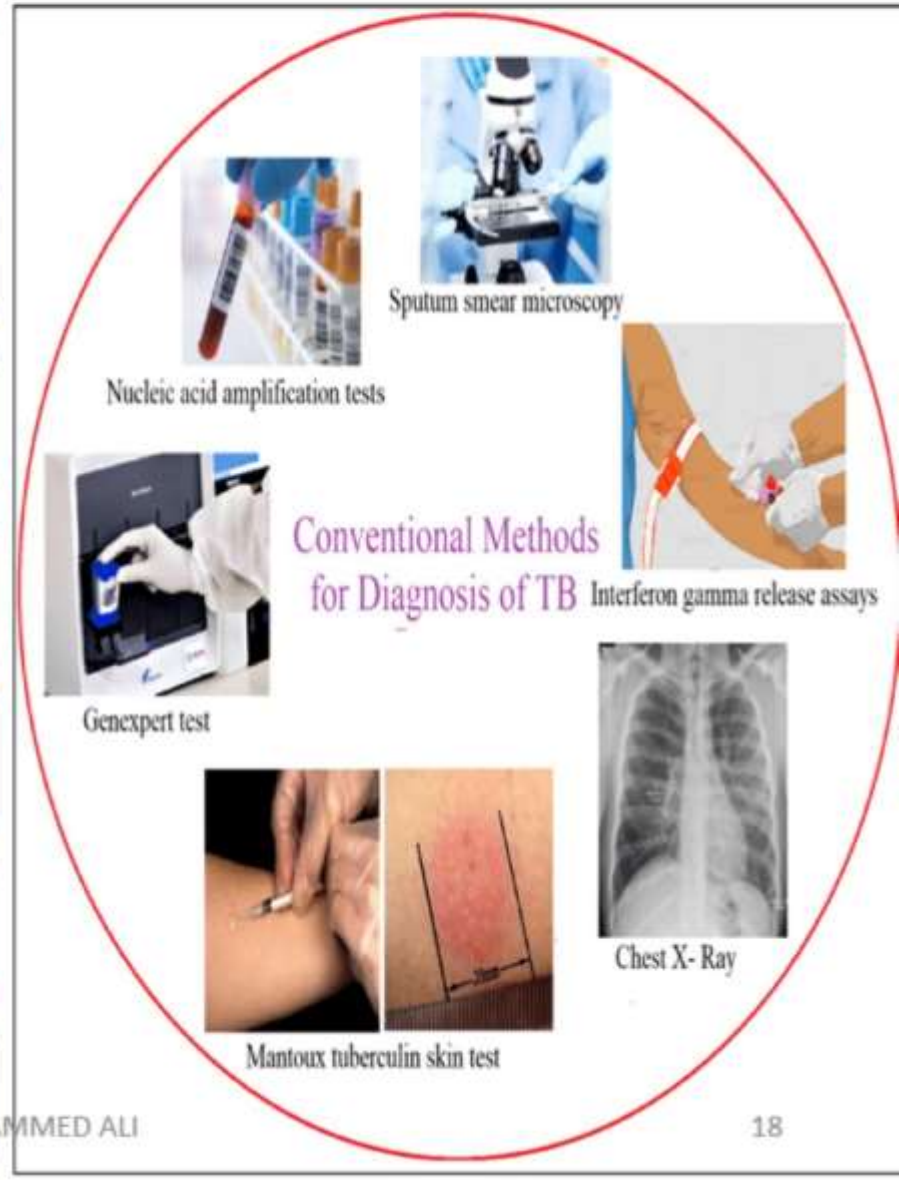
- patient who presents with symptoms or signs suggestive of TB (previously known as TB suspect).

Definite TB Case

- A patient with Mycobacterium tuberculosis complex identified from a clinical specimen, either by culture or by a newer method such as molecular line probe assay.

TB DIAGNOSTICS METHODS

- Sputum collection & Microscopy
- Culture-based diagnostics
- Molecular detection
- Cellular response: identify latent infection and progression to active TB
- Breath biomarker detection
- Antigen-antibody & biomarker detection
- Radiology



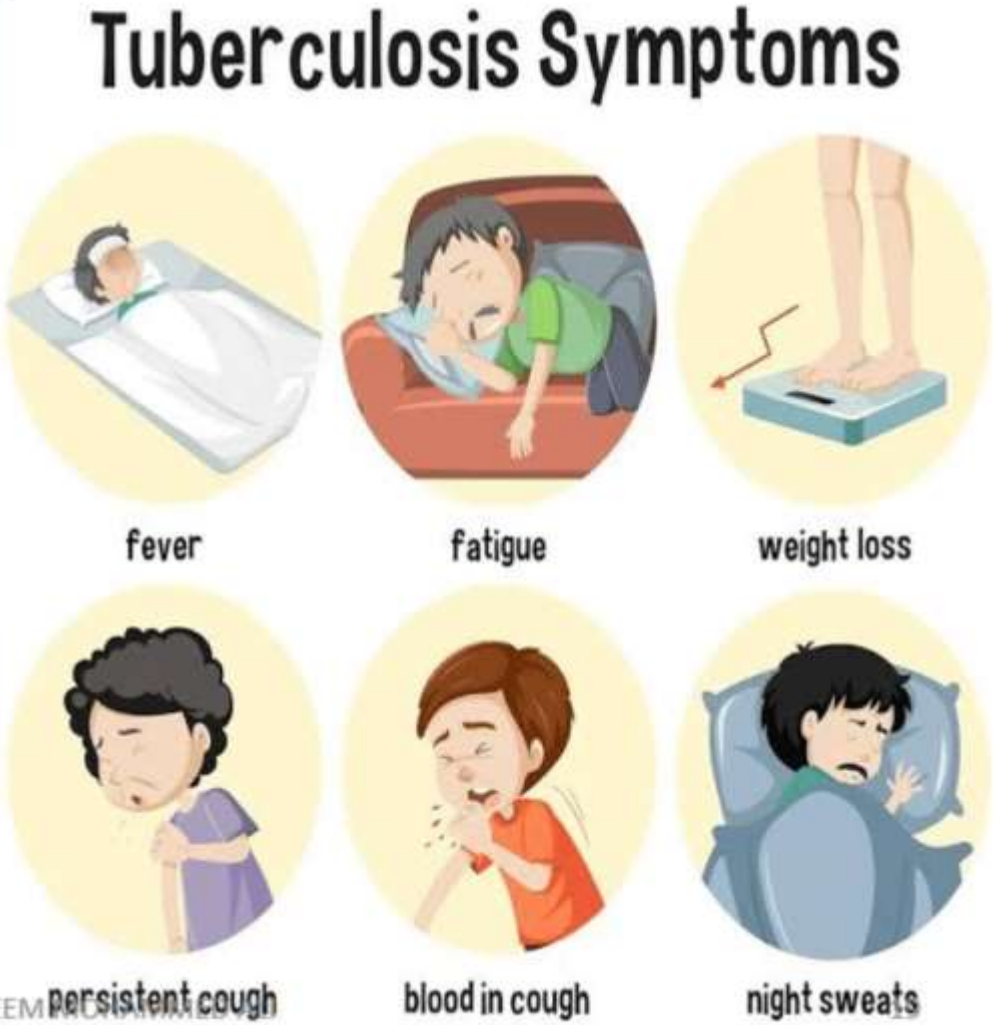
PULMONARY & EXTRA-PULMONARY TB CLINICAL PICTURE

Pulmonary TB:

- Productive, prolonged cough >3 weeks
- Chest pain, hemoptysis
- Systemic: fever, chills, night sweats, fatigue, weight loss, loss of appetite

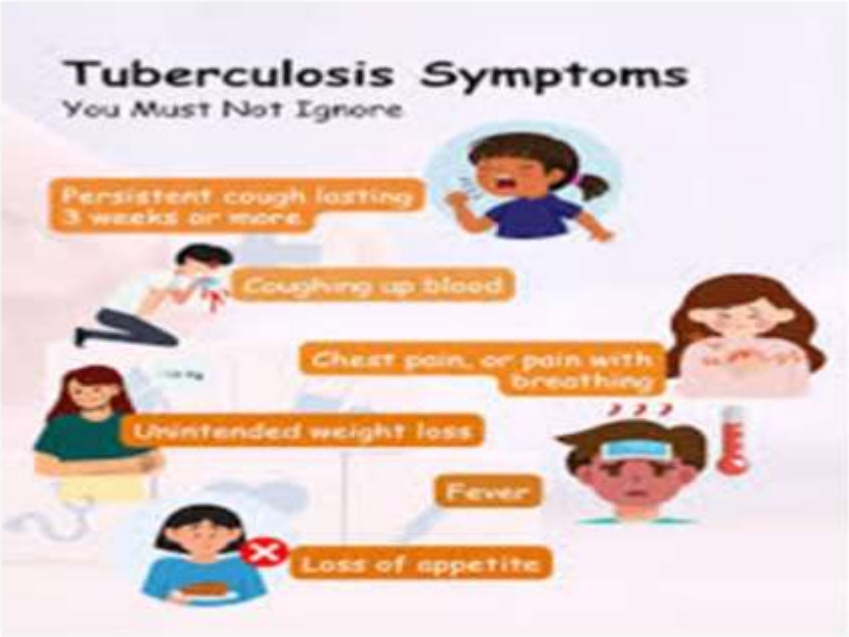
Extra-pulmonary TB:

- Symptoms vary by site:
  - Spine → back pain
  - Kidney → hematuria



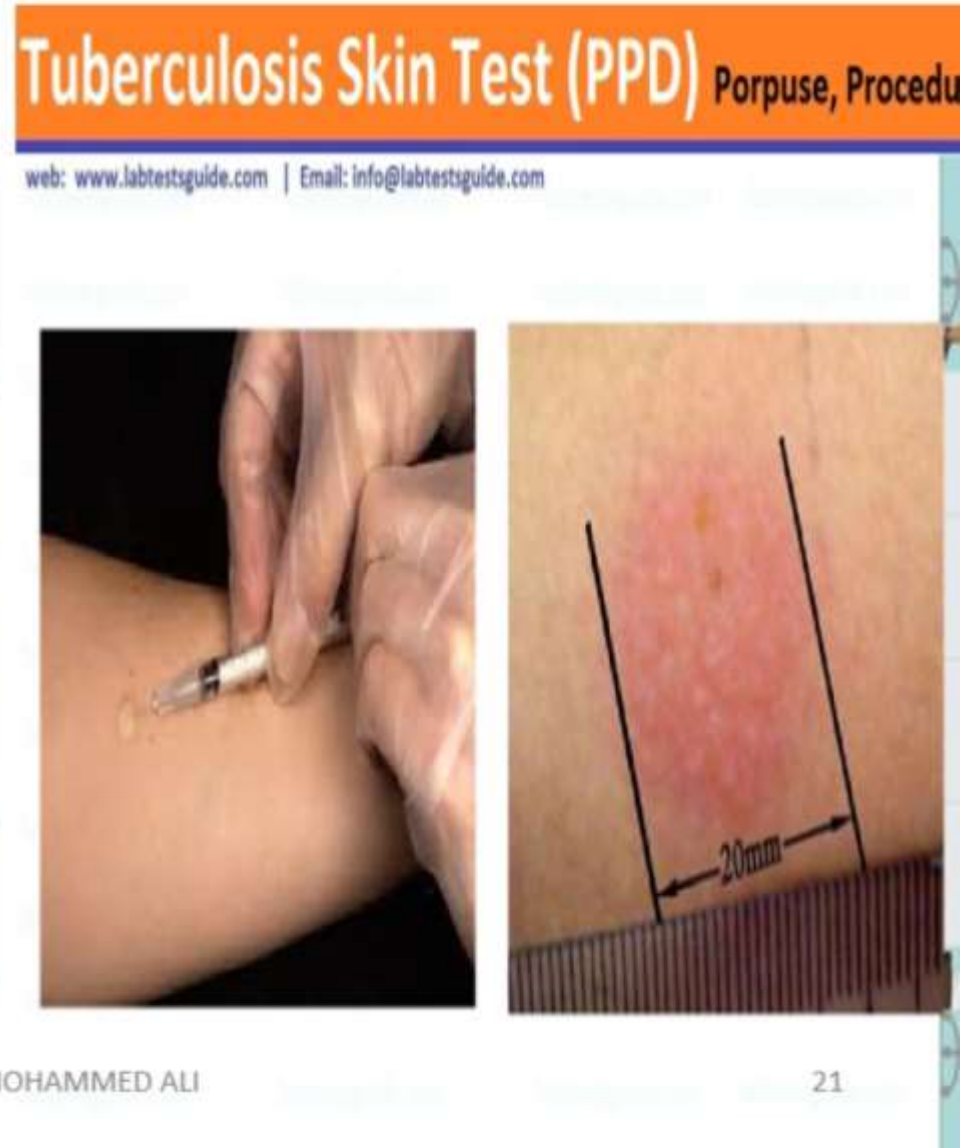
Physical Examination

- Cannot confirm or rule out TB
- Provides information on:
  - Patient's overall condition
  - Factors affecting TB treatment



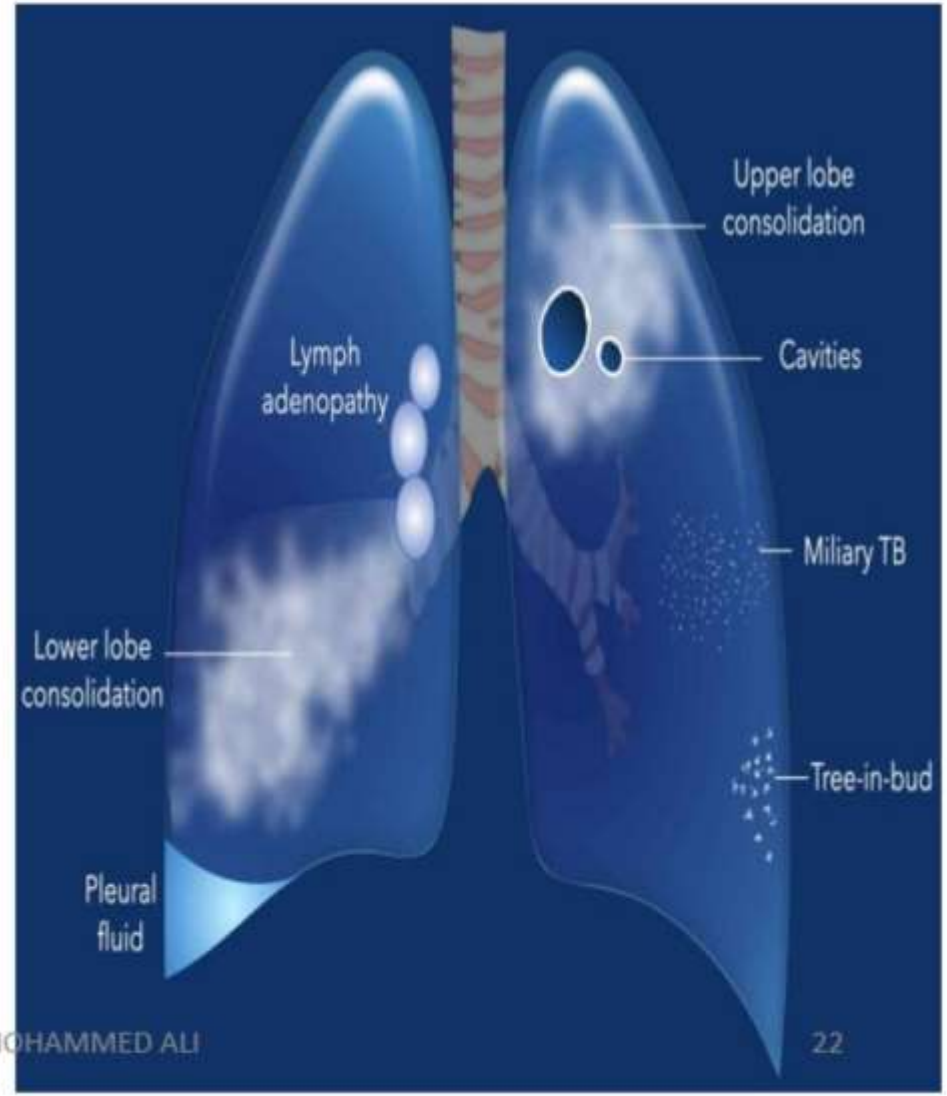
Tuberculin Skin Testing (TST)

- Detect infection in asymptomatic individuals
- Identify TB exposure in groups
- Only way to detect TB infection before progression to TB disease
- ~5–10% of infected people develop active TB



Radiographic Examination

A postero-anterior view of the chest is the standard radiograph needed for the detection and description of chest abnormalities. In some instances, other views (e.g., lateral) or additional studies (e.g., CT scans) may be necessary



LABORATORY EXAMINATION

Detection of acid-fast bacilli (AFB) in stained smears examined microscopically may provide the first bacteriologic clue of TB. (definitive diagnosis)

Diagnosis of Tuberculosis

Methods and Clinical Features of TB Detection ★

1 TB DIAGNOSTIC METHODS

- ✓ **Sputum Collection & Microscopy**
- ✓ Culture-based Diagnostics
- ✓ Molecular Detection
- ✓ **Cellular Response** *(Identify latent TB and progression to active TB)*
- ✓ **Breath Biomarker Detection**
- ✓ Antigen Antibody & Biomarker Detection
- ✓ Radiology

2 CLINICAL FEATURES OF PULMONARY TB

- ✓ **Persistent cough ≥ 3 weeks ± blood in sputum**
- ✓ **Chest pain**, difficulty breathing
- ✓ **Systemic symptoms:**
  - ✓ Fever, chills, night sweats, **weight loss**
  - ✓ Fatigue

*Approximately **15%** of cases are extra-pulmonary*

3 TUBERCULIN SKIN TESTING (TST)

- ✓ Detect infection (without symptoms)
- ✓ Determine TB exposure in groups
- ✓ 5–10% infected develop TB disease

4 RADIOGRAPHIC EXAMINATION

- ✓ Standard chest X-ray (PA view)
- ✓ Other views or CT scan as needed

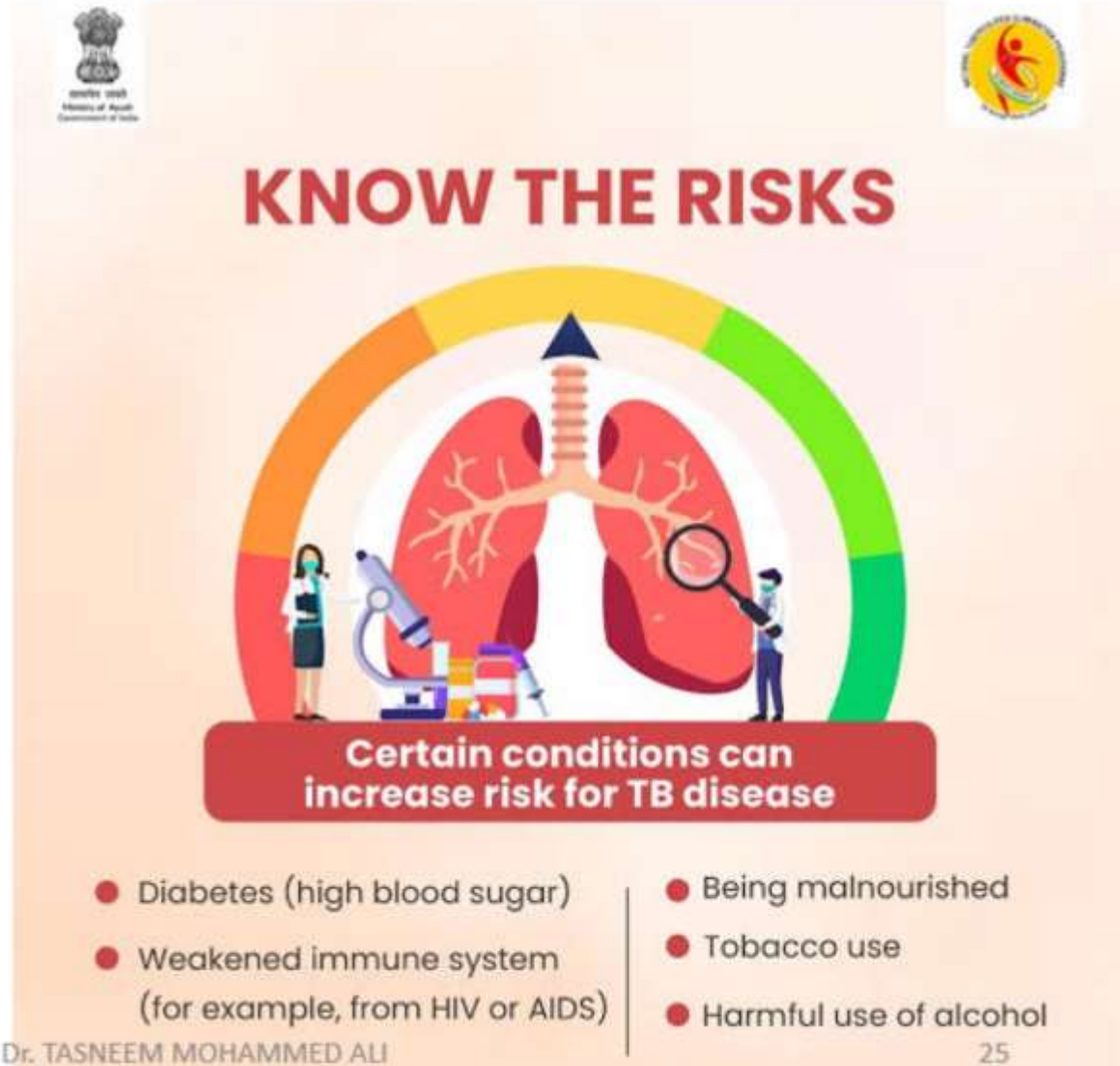
5 LABORATORY EXAMINATION

- ✓ Detection of AFB in stained smears
- ✓ Microscopic examination of sputum smears

***Physical examination** does not confirm **TB** but provides valuable information about the patient's overall condition and risk factors.*

WHO IS AT HIGHER RISK?

- Lifetime risk of TB: 5–10% for infected individuals
- Higher risk in people with:
  - HIV
  - Malnutrition
  - Diabetes
  - Tobacco use



Epidemiologic Determinants of Tuberculosis

Agent Factors

Myobacterium TB
(Koch identified in 1882)

re 3 types that are pathogenic to man:

Human Type

Causes nearly all pulmonary TB cases & 20-50% of extra-pulmonary cases

Bovine Type

Cattle reservoir causes the majority of extra-pulmonary cases (50-80%) & some pulmonary cases

Avian Type

Poultry reservoir rarely infects man causing sporadic extra-pulmonary cases

Risk of clinical TB is high within 6-12 months of infection

Host & Social Factors

Greatest risk in children <3 years, adolescents, young adults, the very old, & immunocomprimed

Risk of clinical TB is high within 6-12 months of infection

Mode of Transmission

TB is airborne, spread via **droplet nuclei** (1-5 microns) expelled by infected person's cough, sneeze, or **talking**

Close contacts are at **highest risk**
(family, roommates, friends, coworkers)

Incubation Period

3-6 Weeks
Since Infection

3-6 weeks Since Infection

1. **Infertucent** of exposure
2. **Environment** of exposure
3. **Duration** of exposure

Progressive TB risk greatest in first 2 years but latent infection can last a lifetime.

Progressive **treatment** can last a **lifetime**

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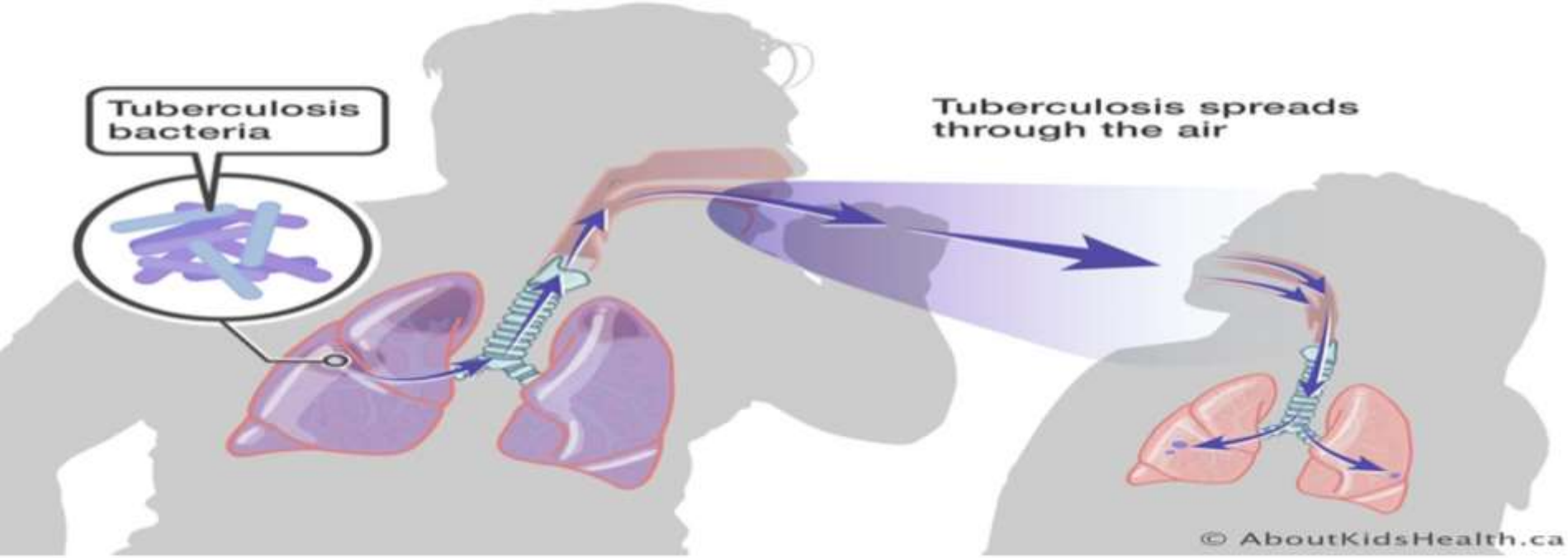
Mode of transmission



When a person with pulmonary or laryngeal TB coughs or sneezes, droplet nuclei containing M. tuberculosis are expelled into the air. These tiny particles (1-5 microns in diameter) can remain suspended in the air for several hours. If another person inhales air containing droplet nuclei, transmission may occur.

The probability that TB will be transmitted depends on three factors

1- the infectiousness of the person with TB

2- the environment in which exposure occurred

3- the duration of exposure.

- Persons at the highest risk of becoming infected with M. tuberculosis are close contacts persons who often spend time with someone who has infectious TB.
- Close contacts may be family members, roommates, friends, coworkers, or others. These persons are at risk for TB infection because they are more likely to be exposed to TB.
- Infection rates have been relatively stable since 1987, ranging from 21% to 23% for the contacts of infectious TB patients.

TUBERCULOSIS

HOST AND SOCIAL FACTORS

TRANSMISSION FACTORS

Those **UNDER 3, ADOLESCENTS, YOUNG ADULTS, VERY OLD** and **IMMUNOSUPPRESSED** are at **HIGHEST RISK**

RISK OF INFECTION

Directly related to **DEGREE OF EXPOSURE**

No significant link to **GENETICS** or **HOST FACTORS**

HEST RISK

6-12 months

DEVELOPMENT HAZARD PERIOD

First 6–12 months after infection

Most hazardous period for **CLINICAL DISEASE DEVELOPMENT**

HIGHEST RISK

6–12 months

RISK OF DISEASE PROGRESSION

Highest in **DEMOGRAPHIC EXTREMES**

Infants → Children → Adolescents → Young Adults

HIGHEST in **3>INFANTS** **LOW** → **HIGH** → Immunosuppressed

Tuberculosis prevention

- Educate the public in mode of spread and methods of control, and the importance of early diagnosis.
- Reduce or eliminate those social conditions that increase the risk of infection, such as overcrowding.
- Use preventive treatment with isoniazid.
- Provide public health nursing and outreach services for direct supervision of patient therapy, and arrange for the examination and preventive treatment of contacts.
- Persons infected with HIV should be skin tested by the Mantoux method, using intermediate strength PPD started on prophylactic treatment if they are PPD positive.

Tuberculosis prevention

- BCG immunization of uninfected (tuberculin negative) people can induce tuberculin reactivity in more than 90% of vaccines. The protection conferred has varied markedly, that protection may persist for as long as 20 years in high incidence situations, while others have shown no protection at all.
- Eliminate bovine tuberculosis among dairy cattle by tuberculin testing and slaughter of reactors; pasteurize or boil milk.
- Take measures to prevent silicosis among those working in industrial plants and mines.

PREVENTIVE THERAPY

preventing the progression of latent TB infection to TB disease.

Standard Regimens: Isoniazid is normally used alone for preventive therapy in a single daily dose of 300 mg in adults and 10 to 15 mg/kg body weight in children, not to exceed 300 mg per dose.

For persons who are at especially high risk for TB and whose adherence is questionable, directly observed preventive therapy should be considered. This method of treatment is especially appropriate in institutions and facilities where preventive therapy can be observed by a staff member.

PREVENTIVE THERAPY

Isoniazid can be given two times a week at a dosage of 15 mg/kg as directly observed preventive therapy.

Before preventive therapy is started, clinicians should:

Rule out the possibility of TB disease (with a medical history and chest radiograph).

Ask for a history of treatment for TB infection or disease.

Aspect	Preventive Regimen (Chemoprophylaxis)	Curative Regimen (Treatment)
Definition	Treatment given to individuals with **latent TB infection** to prevent development of active disease	Treatment given to patients with **active tuberculosis disease**
Purpose	Prevent progression from latent infection to active TB	Cure the patient and stop transmission
Target Group	Close contacts, children, HIV-positive individuals, persons with positive TST but no active disease	Confirmed TB patients (pulmonary or extra-pulmonary)
Number of Drugs	Usually **single drug therapy**	**Multiple drug therapy (combination)**
		Isoniazid (INH),

Aspect	Preventive Regimen (Chemoprophylaxis)	Curative Regimen (Treatment)
Common Drugs Used	Isoniazid (INH)	Isoniazid (INH), Rifampicin (RIF), Pyrazinamide (PZA), Ethambutol (EMB)
Duration	Usually **6–9 months**	Usually **6 months** (2 months intensive + 4 months continuation)
Infectivity of Patient	Person is **not infectious**	Patient may be **infectious**
Public Health Goal	Prevent new TB cases	Treat existing cases and **interrupt transmission**
Supervision	Usually self-administered	Often given under **Directly Observed Therapy (DOT)** recommended by the World Health Organization

Treatment of Tuberculosis Cases

Guidelines for Effective TB Care ★

1 IDENTIFY, DIAGNOSE & TREAT INFECTIOUS PATIENTS

- ✓ Promptly identify, diagnose and treat potentially infectious TB patients
- ✓ Establish case finding and treatment facilities
- ✓ Reduce transmission by treating infectious cases

Without commitment → No sustainable TB control

2 PROVIDE ESSENTIAL MEDICAL FACILITIES

- ✓ Make available medical, laboratory and x-ray facilities
- ✓ Prompt examination of patients, contacts, and suspects
- ✓ Early treatment of high risk individuals
- ✓ Beds for those needing hospitalization

Beds for those needing hospitalization

3 DIRECTLY OBSERVED THERAPY (DOT)

- ✓ Highly effective, directly supervised TB treatment
- ✓ Use appropriate anti-microbial drug combinations
- ✓ Follow standard 6-month regimen:
  - INH + RIF + PZA for 2 months
  - INH + RIF for 4 months

  (+ EMB for areas with high *INH resistance*)

4 HOSPITALIZE SEVERE CASES WHEN NECESSARY

- ✓ Hospital treatment only for severe illnesses
- ✓ Manage those unable to be treated at home

Establish comprehensive case finding, diagnosis, and treatment facilities to effectively manage TB cases

DOTS Strategy to Control TB

The most effective strategy available for controlling the TB epidemic today.

- **Produces cure rates of up to 95%** even in the poorest countries.
- **Prevents new infections** by curing infectious patients.
- **Prevents multidrug-resistant TB (MDR-TB)** by ensuring full course of treatment.
- **Costs as low as $11** for a six-month supply of drugs per patient.
- Ranked by the **World Bank** as one of the "most cost-effective of all health interventions."

Since 1995, Over 10 Million Infectious Patients Treated Successfuly with DOTS

The Five Elements of the DOTS Strategy

DOTS has FIVE key components:

Government Commitment

Government commitment to sustained TB control activities.

Case Detection

Case detection by sputum smear microscopy among symptomatic patients self-reporting to health services.

Drug Supply

Regular, uninterripted supply of all essential anti-TB drugs.

Standardized Treatment Regimen

Six to eight month standardized treat-ment regimen for smear-positive ca-ses, with DOT for last the 2 inial.

Directly Observed Therapy (DOT)

DOTE involves a healthcare **provider directly** observing patients taking their anti-TB medi-cation to ensure adherence and full course completion.

- DOTS prevents: – New infections by curing infectious patients.
- MDR-TB by ensuring full course completion.

TB CONTROL STRATEGY

THE FIVE KEY COMPONENTS OF DOTS ★

(Recommended by the World Health Organization)

1 GOVERNMENT COMMITMENT

Strong Political & Financial Support

- ✓ National TB control policy
- ✓ Sustainable funding
- ✓ Organized TB program
- ✓ Trained healthcare workforce

Without commitment → No sustainable TB control

2 EARLY CASE DETECTION

Sputum Smear Microscopy

- ✓ Cough ≥ 2 weeks
- ✓ Examine sputum samples
- ✓ Detect infectious pulmonary TB

Early detection = Reduced transmission

3 STANDARDIZED TREATMENT + DOT

Directly Observed Therapy (DOT)

- ✓ 6–8 months regimen
- ✓ Direct supervision
- ✓ Ensures adherence
- ✓ Prevents drug resistance

Observed treatment = Higher cure rates

4 UNINTERRUPTED DRUG SUPPLY

Continuous Availability of Quality Drugs

- ✓ No stock-outs
- ✓ Proper drug storage
- ✓ Reliable supply chain

Drug interruption = Treatment failure

5 STANDARDIZED RECORDING & REPORTING

Monitoring & Evaluation System

- ✓ Register every TB patient
- ✓ Track treatment outcomes
- ✓ Measure cure rates
- ✓ Evaluate program performance

DOTS = Detect → Observe → Treat → Supply → Monitor

Reduced Transmission • Increased Cure • Prevented Resistance

Contact Management of TB Case

PPD Testing → All Household Members & Close Contacts

PPD Test Negative

Repeat Skin Test in 2–3 months after exposure ends

Monitor for symptoms of TB

PPD Test Positive

Chest Radiograph (CXR) → Mandatory

Evaluate for Preventive Treatment (PT)

Indicated for :

- Positive reactors (PPD+)
- Initially negative but high-risk contacts:
  - ≤ 5 years old
  - HIV infected
  - Close contacts

Continue PT until:

- Repeat skin test remains negative

Goal: Prevent Progression to Active TB Disease

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TB CASE CONTACT MANAGEMENT

PPD Testing for Household & Close Contacts

PPD Skin Test for all household members and close contacts.

NEGATIVE

Repeat PPD in 2-3 Months After Exposure Ends

POSITIVE

- Chest X-Ray

Preventive Treatment

High-Risk Groups

- Young Children (≤ 5 Years Old)
- HIV Infected Contacts

Continue Treatment Until Repeat PPD is Negative

SUMMARY & IMPORTANT NOTES

TUBERCULOSIS

KEY PUBLIC HEALTH CONCEPTS

TRANSMISSION FACTORS

INFECTIOUS CASE

POOR VENTILATION

LONG EXPOSURE

↑ HIGHER TRANSMISSION RISK

RAPID PROGRESSION IN IMMUNOCOMPROMISED

HIV + WEAK IMMUNITY

PPD RETESTING IMPORTANCE

Detects DELAYED response

DOTS THERAPY SUPERVISION

- ✓ DIRECTLY OBSERVED THERAPY
- ✓ ENSURE ADHERENCE
- ✓ REDUCE RESISTANCE
- ✓ IMPROVE CURE RATE

LATENT TB WITH POSITIVE PPD

PPD +

→ PREVENTS ACTIVE TB

ANY
QUESTIONS?